

VARUN DEEP SAINI

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EDUCATIONS

Year	Degree	Institute
Aug 2026	Bachelors (Computer Science)	Birla Institute of Technology And Science, Pilani

WORK EXPERIENCES

Rippling | Internship Nov 2025 - Present
Software Engineering Intern Bangalore, Karnataka

- Merged 300+ legacy fields (SSNs, National IDs, Tax IDs) across 180+ countries into 3 unified fields with a **controlled rollout**.
- Completed migration of 100+ Standard Global Payroll fields, unblocking new releases in 5+ countries.
- Reviewed 40+ PRs** and **added tests and CI checks** to harden the migration pipeline and **prevent regressions**.

Technologies / Skills Used : Python, Django, Kubernetes, AWS, MongoDB

FamApp (YC S19) | Internship Aug 2025 - Nov 2025
Backend Engineering Intern Bangalore, Karnataka

- Built a high-throughput SMS **data pipeline** exporting **Parquet** batches to **AWS S3** for analytics, processing **450 GB/day**.
- Built a **CDC pipeline** using **Lambda**, **DynamoDB Streams**, and **Kafka** to update PostgreSQL search indices with **sub-second latency**.
- Backfilled **1 Billion** + records using **PostgreSQL cursors** and **batched transactions** without overloading production workloads.

Technologies / Skills Used : Go, gRPC, Kubernetes, AWS, PostgreSQL

Zomato | Internship Jan 2025 - May 2025
Software Engineering Intern Gurgaon, Haryana

- Built a **caching layer** using **Redis**, **Kafka**, and **DynamoDB**, reducing **database read load by 75%**.
- Achieved **6x Redis cost reduction** via Protobuf serialization and Snappy compression (**payload: 15KB to 3KB**).
- Integrated Gold surge fee segmentation, separating Gold vs. Non-Gold pricing, **adding 10p profit per order**.
- Optimized service performance** by introducing **local caching and polling**, which eliminated an RPC call, **reduced latency by 30ms**, and **removed dependency on an external service**, leading to a **25% reduction in its container count**

Technologies / Skills Used : Go, gRPC, Microservices, Redis, Kafka

Allen Digital | Internship Sep 2024 - Dec 2024
Software Engineering Intern Bangalore, Karnataka

- Optimized distributed FFmpeg transcoding: **10–15x faster** processing, **4x compression gains**, saving **\$360K/year in infra costs**.
- Built a Rust-based **GStreamer pipeline** with Mediasoup SFU to upload **chunked MP4s** to **GCP Storage** fault-tolerantly.
- Automated **stitching and synchronization** of participant video/audio streams into final MP4s, **owning end-to-end pipeline**.

Technologies / Skills Used : Go, Rust, Mediasoup, FFmpeg

AWARDS & ACHIEVEMENTS

- Candidate Master** @Codeforces - Max Rating 1953 and 5* @CodeChef - Max Rating 2125.
- Qualified for ICPC Regionals twice (2023, 2024), secured Rank 75 at 2023 ICPC Chennai Asia West Regionals.
- Secured World Ranks 542 and 501 in Meta Hacker Cup Round 2 (2024, 2025), missed semifinals by **1 rank** in 2025.
- Won the **Silver Medal** (Second Place) at **Rubrik Go For Gold ICPC Camp 2024**.
- Achieved **All India Rank 12** in Codeforces Round 976 and **World Rank 8** in CodeChef Starters 164 - Div 1.

OPEN SOURCE CONTRIBUTIONS

MariaDB Server C, SQL

- Implemented the `UUID_TIMESTAMP()` function in MariaDB to extract timestamps from UUIDv1/v7 values. (#4496)
- Added per-session deadlock diagnostics to InnoDB, surfacing deadlock details via `SHOW WARNINGS`. (#4492)

Databricks Asset Bundles (DABs) - CLI Go

- Implemented Write-Ahead Log (WAL) for direct deployments, preventing orphaned resources on deploy crashes. (#4106)
- Fixed dependency-order deletion in the direct deployment engine, resolving failures from deleting dependencies before their dependents. (#4105)
- Built KillCaller test server feature to simulate CLI process crashes, enabling testing of crash scenarios. (#4146)

PROJECTS

Ultra Fast Key-Value Cache | Github Feb 2025

- Built a **sharded in-memory KV store** using djb2 hashing; pre-allocated **700K entries/shard** to eliminate GC pauses.
- Applied **kernel-level TCP tuning** and fasthttp for **zero-allocation handling**; containerized with **1M+ file descriptor support**.

Technologies / Tools Used : Go, TCP, Threads, Semaphores

SKILLS

Programming Languages : Go, C++, Python, Java, JavaScript, Rust, SQL, C, TypeScript
Frameworks & Libraries : gRPC, REST, Microservices, Redis, Kafka, Docker, Kubernetes, FastAPI, Next.js, Spring Boot
Tools & Platforms : AWS, GCP, Linux, CI/CD, S3, Docker, Kubernetes, Jenkins, Git, Temporal
Databases : PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, DynamoDB